Himanshu Singhal

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Education

Indian Institute of Technology Guwahati, India — Oct 2022 - Present
Bachelors of Technology in Data Science and Artificial Intelligence — GPA: 8.10/10.0

Experience

Adobe Research — May 2025 - Aug 2025
Summer Intern, Internship — Bangalore, India

- Worked on **enterprise-branded asset generation** using **precise trademark colors** with Diffusion Transformers.
- Curated a **plug-and-play mapper** that translates **hex codes into embeddings** for **precise colorization** via prompts.
- Improved generated color accuracy by **64%** over baseline methods across a self-curated **1200 prompt benchmark**.
- Documented and submitted a **patent application** detailing the end-to-end mapper architecture;the application was reviewed, **approved, and accepted** through Adobe's internal patent review process.

Indian Institute of Technology, Guwahati — Jul 2025 - Present
BTech Project, Prof. Suresh Sundaram — Guwahati, India

- Investigating **online signature verification** using **Vision Transformers** to better model temporal dependencies and represent fine-grained signature dynamics, leading to improved authentication accuracy.
- Reformulated signature verification as a **one-class anomaly detection** problem, allowing the model to be trained solely on genuine samples without requiring forged data for supervision.
- Modeled temporal signature signals as **image-based representations** using **Gramian Angular Summation Field (GASF)** plots to capture fine-grained dynamics while mitigating temporal distortions and local misalignments.
- Evaluated the proposed framework on four public benchmarks—**BioSecureID, SVC2021, DeepSignDB, and MCYT-100**—achieving performance competitive with state-of-the-art methods.

Arizona State University - CORAL Lab — Feb 2025 - Present
Research Internship, Prof. Vivek Gupta — Remote

- Working on the problem of multi-table SQL generation, focusing on accurately generating queries that span multiple tables while **avoiding the use of LLMs** for table selection.
- Developing an **end-to-end SQL generation pipeline** that leverages **inter-table join relationships** for intelligence table selection and pruning with **multiple required tables**, without using LLMs for pruning and table selection.
- Achieved performance comparable to state-of-the-art LLM-based approaches, demonstrating substantial efficiency gains through a **92.24%** reduction in token cost and up to **4.6×** improvement in inference latency.
- Submitted to **ACL 2026**; received a strong **meta-review score of 4/5** and awaiting final acceptance decision. **Paper**: arXiv:2511.00805

Indian Institute of Technology, Guwahati — Jan 2025 - May 2025
Research Intern, Prof. Ayon Borthakur — Guwahati, India

- Implemented and evaluated the **Titans neural memory architecture** by integrating it with GPT-2 tokenizer, enabling systematic benchmarking on standard language understanding tasks.
- Adapted and trained Titans under compute constraints using the **WikiText dataset**, and evaluated performance across multiple benchmarks via the **LM-Eval** framework.
- Explored a **hybrid memory–attention design** combining Titans and Hymba, focusing on architectural interactions between persistent memory, attention mechanisms, and sequence modeling.

Devrev Cloud India Private Limited — Feb 2024 - Apr 2024
Software Intern, Internship — Remote

- Developed a **customizable NPS survey snap-in** for developer organizations, enabling them to dispatch surveys at selected intervals, streamlining feedback collection, and boosting user engagement.
- Utilized **hooks** and **cron jobs** to facilitate sending emails at specified intervals, and employed frameworks such as **React-Email** to guarantee compatibility and a consistent user experience across different email clients.
- Built **real-time** survey update mechanisms and **interval-based** email dispatch system ensuring consitency.
- Deployed an automated survey distribution system serving **1,000+ users**, improving feedback collection efficiency.

Publications

REaR: Retrieve, Expand and Refine for Effective Multitable Retrieval. — Under Review (ACL 2026)
R. Agarwal*, **H. Singhal*** P.B. Chen, M.R. Choudhury, D. Roth, V. Gupta

- *Submitted to* **ACL 2026***; received a* **meta-review score of 4/5** *and currently under* review.
- **arXiv***: 2511.00805*

Projects

Agentic RAG Chatbot for LegalQA Oct 2024 - Dec 2024
Inter IIT Tech Meet 13.0 Github

- Solved the problem of **unreliable legal RAG systems** that **hallucinate** and retrieve inaccurate information.
- Implemented **parallel DAG-based** task planning and scheduling by modifying **LLMCompiler** for RAG.
- Incorporated **Pathway's vector store** for **real-time data processing** to handle document **ingestion** and **retrieval**.
- Achieved **86% accuracy** on the **CUAD** dataset (contract law) and **84% accuracy** on the **AILA** dataset (case law).

Adobe Behaviour Simulation Challenge Oct 2023 - Dec 2023
Inter IIT Tech Meet 12.0 Github

- Leveraged classic ML Models, Transformers, and **LLMs** for **behaviour** and **content simulation** on tweets.
- Trained a **DistilBERT** classifier to bucket tweets, followed by separate **bucket regressors** for likes prediction.
- Generated descriptions of media using **BLIP-2** and combined with metadata to finetune **Llama-2** for captioning.
- Achieved a bucket classification accuracy of **74.3%** and **BLEU-1** score of **0.13** with inference times **0.9s** and **5.4s.**

Devrev's AI Agent 007 Oct 2023 - Dec 2023
Inter IIT Tech Meet 12.0 Github

- Engineered a **conversational LLM chatbot** with **dynamic tool selection** capabilities for intelligent query resolution.
- The chatbot selects and sequences **appropriate tools** with required arguments to accurately answer user queries.
- Implemented a pipeline integrating **Retrieval-ICL** for few-shot retrieval, **Chain-of-Thought** prompting, **corrective reprompting**, and **RAG** with **Dynamic Retrieval** to bolster the model's contextual understanding.
- Achieved high-performance pipeline with **sub-10-second query latency** and cost efficiency under **$0.02 per query**.

Other Projects

- **Telematics Data Analysis :** Data pipeline with **real-time Kafka streaming**, **risk prediction**, and **visualisation dashboard**
- **Interview Chatbot :** Chatbot with **Google Vertex AI** for **stimulated interview** and feedback with **Codeforces API**

Achievements

Adobe Research	**Patent application** authored from internship research and **approved** through internal review	2025
Inter IIT Tech Meet 13.0	Secured **Rank 4** among all **23 IITs** in ML/AI Problem Statement by Pathway	2024
Encode, IITG	Team secured **first position** out of **2342** participants at ML hackathon across India	2024
Amazon MLSS	Selected to attend ML lectures from more than **20,000 students in India**.	2024

Technical Skills

Programming: C++, C, Python, JavaScript, TypeScript, R
ML/AI: Pytorch, Numpy, Pandas, TensorFlow*, Langchain, Langgraph, Llamaindex
Database Management MySQL, Hadoop*, Spark, Kafka
Miscellaneous: Git, Shell, Latex, Docker

Relevant Coursework

Computer Science: Data Structures and Algorithms (Theory & Lab), Database Management Systems (Theory & Lab), Computer Systems, Big Data Analytics, Machine Learning, Deep Learning, Data Mining, Advanced Deep Learning
Mathematics: Linear Algebra, Statistical Foundations of Data Science, Applied Probability & Random Processes

Positions of Responsibility

- **Teaching Assistant, DA205**, Responsible for curating weekly exam question banks and supporting live sessions.
- **Secretary, IITG.ai**, Responsible for managing events, workshops, competitions, and discussions at IITG.ai club, the AI Community of IIT Guwahati.